

Merkel Omegat OMS-MR

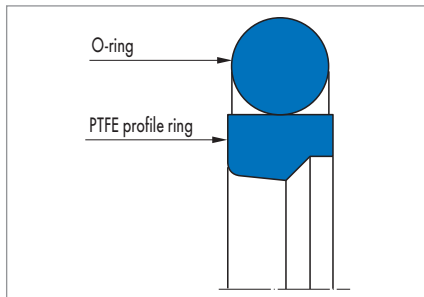


Fig. 1 Merkel Omegat OMS-MR

Product description

Two-piece Merkel seal set made of PTFE profile ring with one O-ring as a pre-load component.

Product advantages

Rod seal used in particular in a sealing system

- Very high resistance to pressure
- Good thermal conductivity
- Very good extrusion resistance
- High resistance to abrasion
- Low friction, free of stick-slip.

Application

Industrial vehicles, handling equipment, agricultural machinery, cranes, injection moulding machines, presses, marine hydraulics, control and regulation equipment, mills.

Material

PTFE profile ring

Material	Code
PTFE bronze compound	PTFE B602
PTFE glass/MoS ₂ compound	PTFE GM201

O-ring

Material	Code	Hardness
NBR	70 NBR B276	70 Shore A
FKM	70 FKM K655	70 Shore A

Operating conditions

Material	PTFE GM201/NBR	PTFE B602/ 70 FKM K655	PTFE B602/ 70 NBR B276
	Temperature range in °C		
Hydraulic oils HL, HLP	-30 ... +100	-10 ... +200	-30 ... +100
HFA fluids	+5 ... +60	–	–
HFB fluids	+5 ... +60	–	–
HFC fluids	-30 ... +60	–	–
HFD fluids	–	-10 ... +200	–
Water	+5 ... +100	–	–
HETG (rapeseed oil)	-30 ... +80	-10 ... +80	-30 ... +80
HEES (synthetic ester)	-30 ... +80	-10 ... +100	-30 ... +80
HEPG (glycol)	-30 ... +60	-10 ... +80	-30 ... +60
Mineral greases	-30 ... +100	-10 ... +200	-30 ... +100
Pressure p in MPa	40		
Running speed v in m/s	5		

Surface quality

Surface roughness	R _a	R _{max}
Sliding surface	0,05 ... 0,3 µm	≤2,5 µm
Groove base	≤1,6 µm	≤6,3 µm
Groove flanks	≤3,0 µm	≤15,0 µm

Percentage contact area M_r >50% to max. 90% at cutting depth c = Rz/2 and reference line C ref = 0%.

Design notes

Please observe our general design notes in → Technical Manual.

Gap dimension

The decisive factor for the function of the seal is the largest gap dimension occurring during operation on the non-pressurised side of the seal. → Technical Manual.

Profile		Max. permissible gap dimension			
L	Profile	16 MPa	26 MPa	32 MPa	40 MPa
2,2	2,45	0,35	0,30	–	–
3,2	3,65	0,40	0,35	–	–
4,2	5,35	0,50	0,40	0,30	–
6,3	7,55	0,55	0,45	0,35	0,30
8,1	10,25	0,60	0,50	0,40	0,40
8,1	12,00	0,70	0,60	0,55	0,50
9,5	13,65	0,75	0,65	0,60	0,55

Tolerance recommendation and dimension D2

The admissible gap width, tolerances, guide play and deflection of the guide under load are to be taken into account when designing D2. → Technical Manual.

Nominal Ø d	16 MPa		26 MPa		32 MPa		40 MPa	
	d	D	d	D	d	D	d	D
≤80	f8	H9	f8	H9	f7	H9	f7	H7
>80 ... 500	f8	H8	f8	H8	f7	H8	f7	H8
>500 ... 1450	f8	H8	f7	H8	f7	H8	f7	H8

The dimensions D1 and DF are to be viewed in connection with the guide elements used.

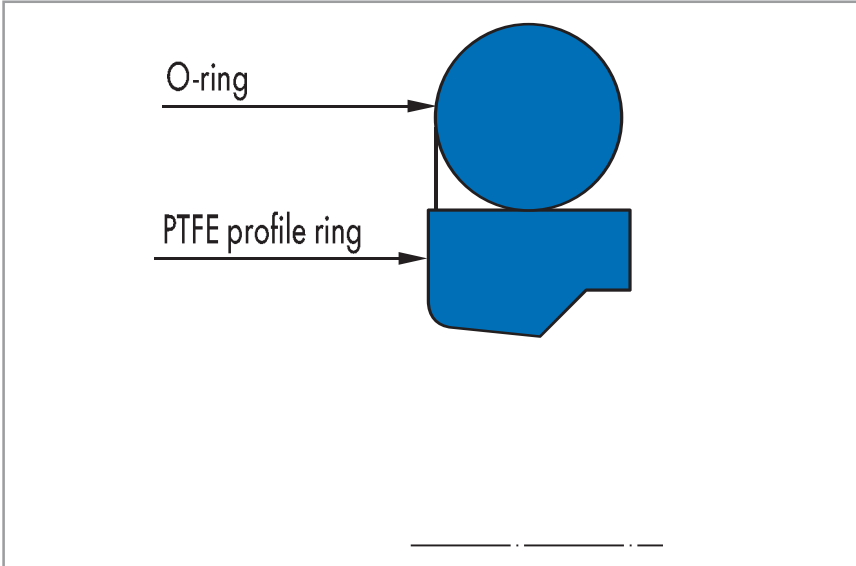
Fitting & installation

Careful fitting is a prerequisite for the correct function of the seal.

→ Technical Manual.



Merkel Omegat OMS-MR



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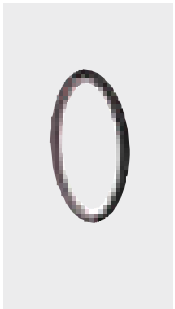
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HFA fluids	+5 ... +60	-	-
HFB fluids	+5 ... +60	-	-
HFC fluids	-30 ... +60	-	-
HFD fluids	-	-10 ... +200	-
Water	+5 ... +100	-	-
HETG (rapeseed oil)	-30 ... +80	-10 ... +80	-30 ... +80
HEES (synthetic ester)	-30 ... +80	-10 ... +100	-30 ... +80
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Profile		Max. permissible gap dimension			
L	Profile	16 MPa	26 MPa	32 MPa	40 MPa
2,2	2,45	0,35	0,30	-	-
3,2	3,65	0,40	0,35	-	-
4,2	5,35	0,50	0,40	0,30	-
6,3	7,55	0,55	0,45	0,35	0,30



Profile		Max. permissible gap dimension			
L	Profile	16 MPa	26 MPa	32 MPa	40 MPa
8,1	10,25	0,60	0,50	0,40	0,40
8,1	12,00	0,70	0,60	0,55	0,50
9,5	13,65	0,75	0,65	0,60	0,55

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The admissible gap width, tolerances, guide play and deflection of the guide under load are to be taken into account when designing D2.

Nominal Ø d	16 MPa		26 MPa		32 MPa		40 MPa	
	d	D	d	D	d	D	d	D
≤80	f8	H9	f8	H9	f7	H9	f7	H7
>80 ... 500	f8	H8	f8	H8	f7	H8	f7	H8
>500 ... 1450	f8	H8	f7	H8	f7	H8	f7	H8

The dimensions D1 and DF are to be viewed in connection with the guide elements used.

Fitting & installation

Careful fitting is a prerequisite for the correct function of the seal.



Installation of Hydraulic Seals

Before installing the sealing components the complete system must be cleaned to remove machining residues, chips, dirt and other particles. Seals must not be pulled over sharp edges, threads, feather key grooves or similar when mounting. These parts must be covered during fitting (→ Fig. 37). Sharp edges must be de-burred or chamfered or radiused. Never use sharp-edged tools. Seal, piston rod and cylinder bore must be oiled or greased before fitting. Heating the seals before installation in +80 °C to +100 °C hot oil will make the seal material more elastic and it will be easier to install the seal.

Insertion chamfers on rods and pipes

To prevent damage to sealing components when mounting, cylinder bores and piston rods must be chamfered. For the surface quality of the chamfer $R_a \leq 4 \mu\text{m}$ is applicable.

The edge at the transition from the chamfer to the sliding surface must be rounded and polished. Product-specific information can be found in the shape descriptions.

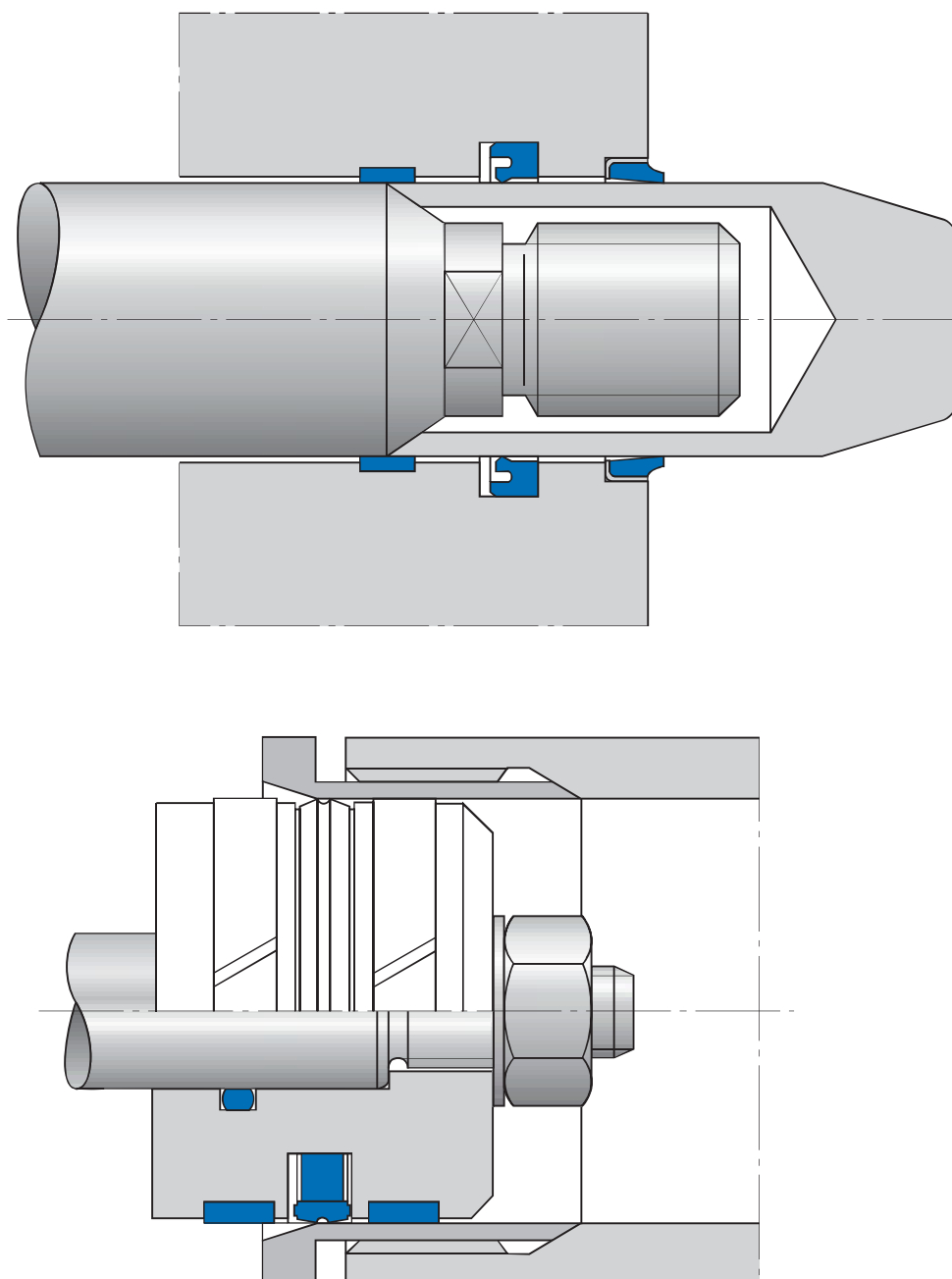


Fig. 37 Covering threads when mounting seals

Fitting of rod seals

When mounting rod seals two types of installation must be distinguished (→ Fig. 38):

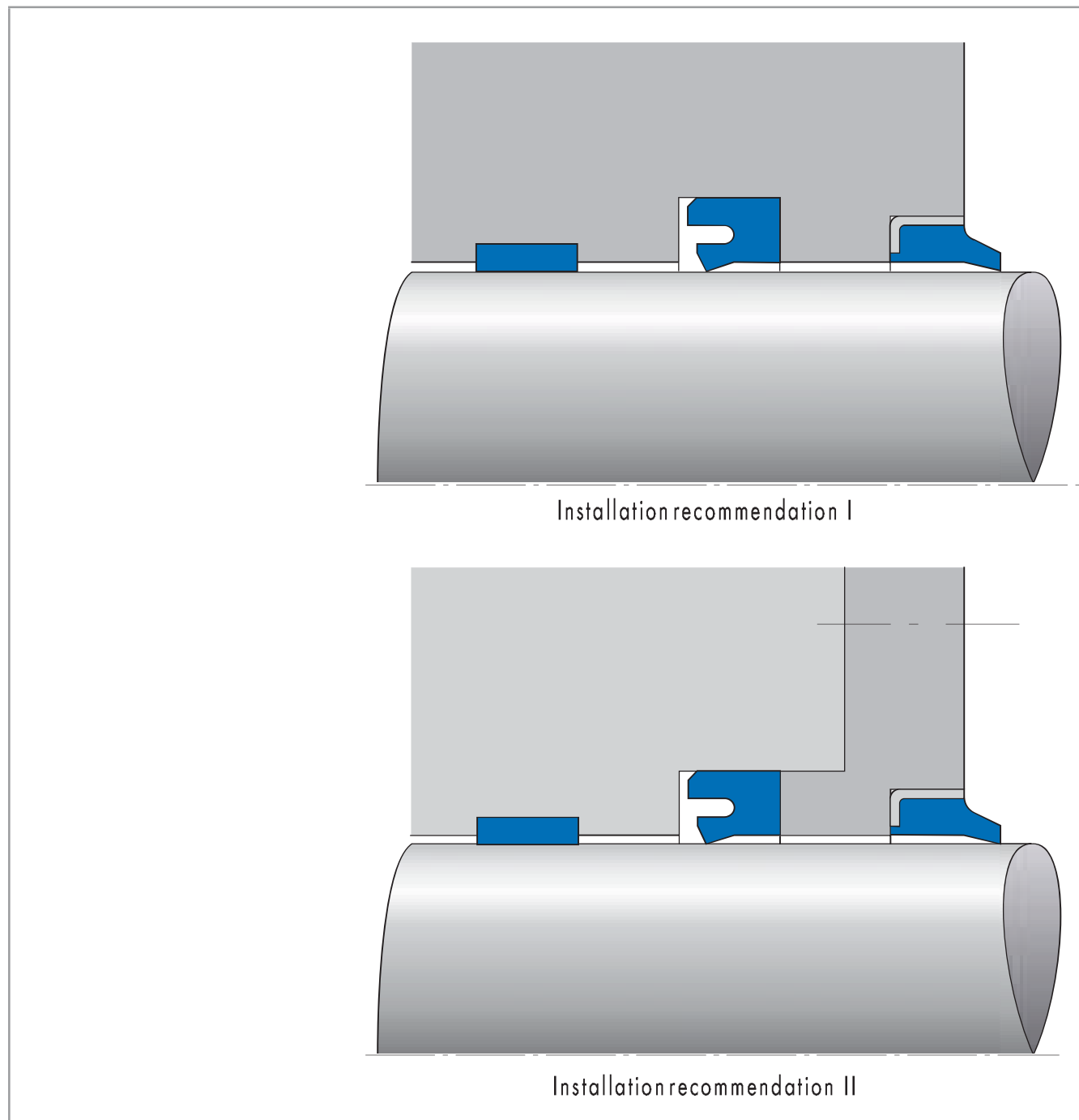


Fig. 38 Installation types of rod seals

- Snap-in fitting in an undivided installation space (installation recommendation I).
Seals that are suitable for this type of installation are marked with h and w in the tables of dimensions.
- Fitting in a divided installation space (installation recommendation II).
Seals that required a divided installation space are not marked in the tables of dimensions.

Assembly tools for rod seals

Snap-in fitting in undivided housings (recommended installation) can be made much easier with suitable assembly tools. The fitting tool I (order no. 00375753) can be used to snap in U-rings from 35 mm diameter

(profile size 5 mm) up to a nominal diameter of 80 (profile size 10 mm) in undivided grooves. The ring is pressed into a kidney shape and pushed into the rod guide. The fitting tool is withdrawn after the seal has snapped into the groove.

Another option for snap-in fitting of rod seals is to use a suitable stopper and a rod (→ Fig. 41).

Here the seal is initially placed manually in the groove and then pushed with a rod until it snaps into the groove. Stopper and rod should be manufactured from a suitable plastic.



Fitting of groove and compact seals with back-up ring
The Merkel SM U-Ring (primary seal) with locked-in back-up ring can be snapped into a plunge-cut groove. The sealing ring is first inserted into the groove. Then the back-up ring is installed.
Compact seals with a locked-in back-up ring can be snapped into a plunge-cut groove, depending on the diameter and profile.

Fitting of multi-part compact seals for the rod: Merkel Omegat OMS-MR
For rod diameters ≤15 mm an axially accessible housing is required. Up to rod diameter 28 mm an axially accessible housing is recommended. If this is not possible for design reasons, the seal must be selected in accordance with the smaller installation size L.
For diameter range 38 to 50 mm we also recommend using the seal in accordance with the smaller fitting size L because of the easier fitting (→ Fig. 41). The max. permissible gap widths of the type must be observed.



Fig. 39 Fitting tool I for rod seals



Fig. 40 Fitting tool II for rod seals

Fitting in divided housings
From a specific nominal diameter, depending on the profile size, rod seals must be installed in divided housings. The limit sizes are listed in d (→ Tbl. 8).

Rod seals in divided housings (fitting recommendation II) can be installed without special tools. For series installation we recommend using mounting sleeves and pilot shafts (→ Fig. 42).

U-rings and single-piece compact seals							
Profile size $P = \frac{D_N - d_N}{2}$	4	5	6	7,5	10	12,5	15
Limit nominal di- ameters for snap-in fitting	25	30	40	50	80	100	105
Seals that are suitable for snap-in fitting are identified with h (hand) in the descriptions in the dimension lists.							

Tbl. 8 Limit dimensions for snap-in fitting (required values)

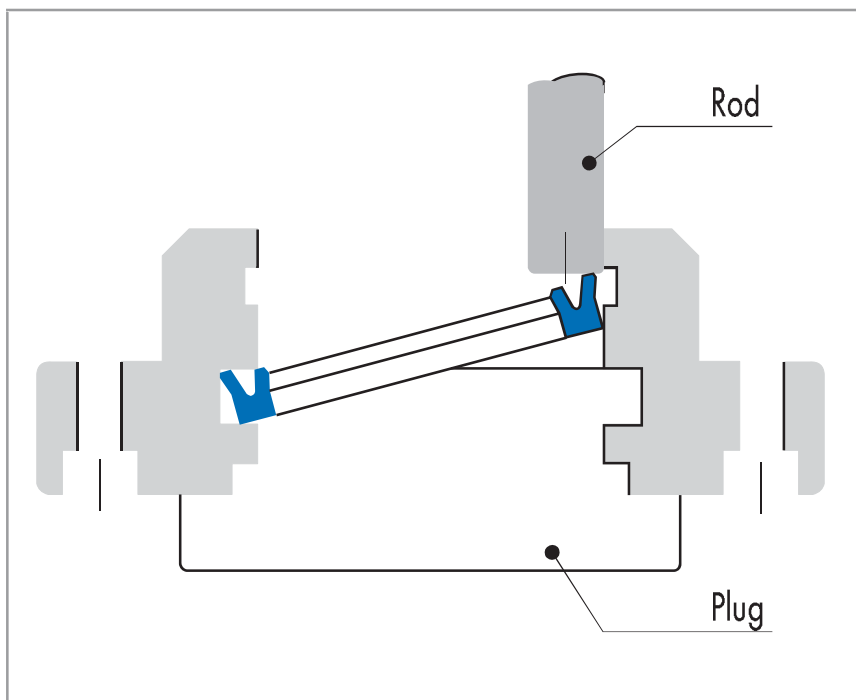


Fig. 41 Installation aid for rod seals

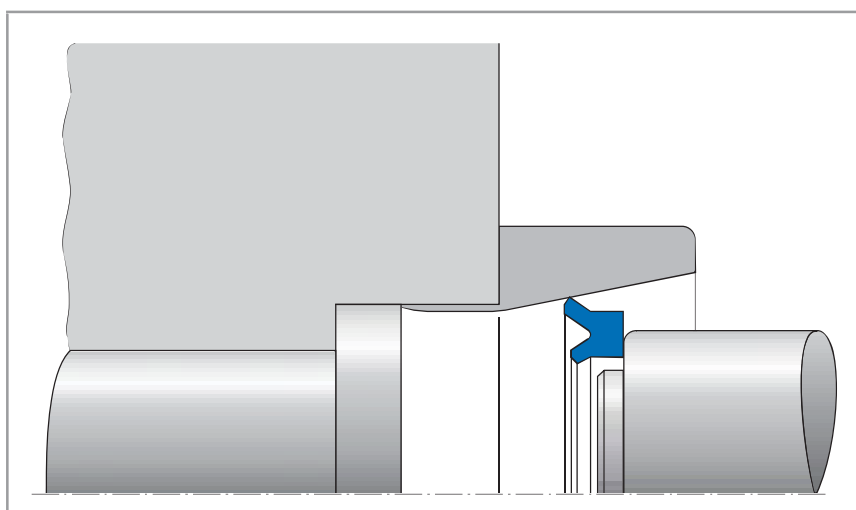


Fig. 42 Fitting of rod seals in divided housings

Fitting of piston seals

Similar to fitting of rod seals there are also two types of installation of piston seals:

- Snap-in fitting in undivided housings.
Seals that are suitable for this type of installation are identified with h or w in the dimension tables.
- Fitting in divided housings.
When assembled the metal parts must adhere to preclude extrusion wear on the static side.

Multi-part compact seals for pistons

Simko Piston Seals can generally be snapped in without tools. The following illustrations show the snap-in fitting without tools for the Merkel Compact Seal Simko 300 Compact Seal. First the elastic rubber contact pressure element is snapped in. Then the PUR sealing ring is placed on one side in the groove and pushed over the previously oiled piston body until it snaps completely into the groove.

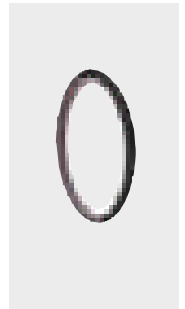


Fig. 43 Snapping in the polyurethane running ring



Fig. 44 Snapping in the polyurethane running ring



Assembly tools for piston seals

The snap-in fitting is made much easier with suitable assembly tools.

Simko Piston Seals as well as U-Rings that in some cases are used as single-acting piston seals can be easily installed with the aid of simple installation tool. See the following illustrations:



Fig. 45 Fully mounted seal



Fig. 47 Inserting mounting pin



Fig. 46 Inserting sealing ring



Fig. 48 Turning mounting lever

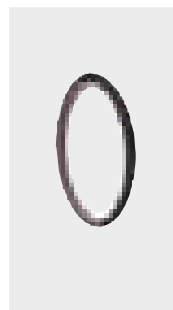


Fig. 49 Snapping in the guide back-up ring

Fitting of compact seals of the Omegat series for pistons and rods

The Merkel Omegat Compact Piston Seals (OMK-MR, OMK-S, OMK-E, OMK-ES) and the Merkel Omegat Compact Rod Seals (OMS-MR, OMS-S) are suitable for undivided housings in virtually all dimensions. Fitting requires special care.

To prevent damage to the sealing edge, which may result in leakage before commissioning, it is important to observe our installation instructions.

Fitting for installation

Omegat Seals consist of a high-quality pressure and wear-resistant profile ring and an O-ring as the pre-load component. Careful fitting is essential for correct function.

Before starting installation, make sure that:

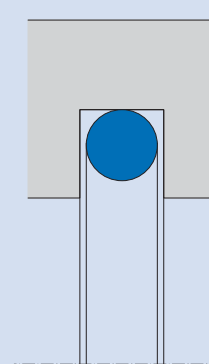
- The required insertion chamfers on the piston rod and cylinder bore have been de-burred and rounded,
- Thread peaks and sharp edges are covered,
- Dust, dirt, chips and other external objects must be thoroughly removed,
- The Omegat sealing components and the components are oiled or greased (use only greases without solid additives. In this regard make sure that they are compatible with the medium.),
- The assembly tools must be of soft material and have no sharp edges.

Heating them in oil up to about 80 °C makes it much easier to stretch and deform the Omegat Profile Ring.

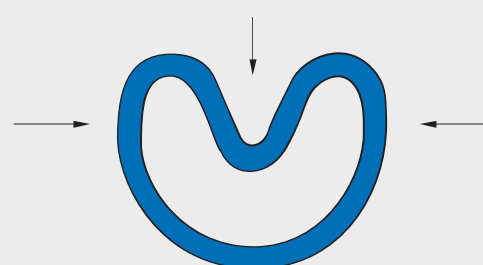
Omegat Rod Seal

Fitting of Omegat Rod Seals in undivided housings is very easy ($d \leq \varnothing 15$ axially accessible housing required):

Insert O-ring into the groove without twisting it.



Press Omegat Profile Ring into a kidney shape (attention: do not kink!).





Omegat Rod Seal

A fitting tool is preferred for series fitting.

Place compressed Omegat Profile Ring on the O-ring so the sealing edge is on the pressure side.

Insert Omegat Profile Ring in original shape in the groove.

Then calibrate with a mandrel. The mandrel can be manufactured from PA, POM or similar material. Chamfers of 15° and minimum 30 mm long.

Direction of pressure P

Tbl. 9 Deformation of the profile ring

Ø d	L	Ø d mandrel
<50	15	Ø d – 0,10
≥ 50 ... <120	20	Ø d – 0,18
≥120 ... <200	30	Ø d – 0,25
≥200 ... <650	40	Ø d – 0,35
≥650 ... <900	50	Ø d – 0,50

Tbl. 10 Recommended diameter of mounting equipment

Recommendation:

Please use an assembly tool when $d \varnothing > 15$ mm and for larger series. This deforms the profile ring less. The major design principles are shown in the drawing.

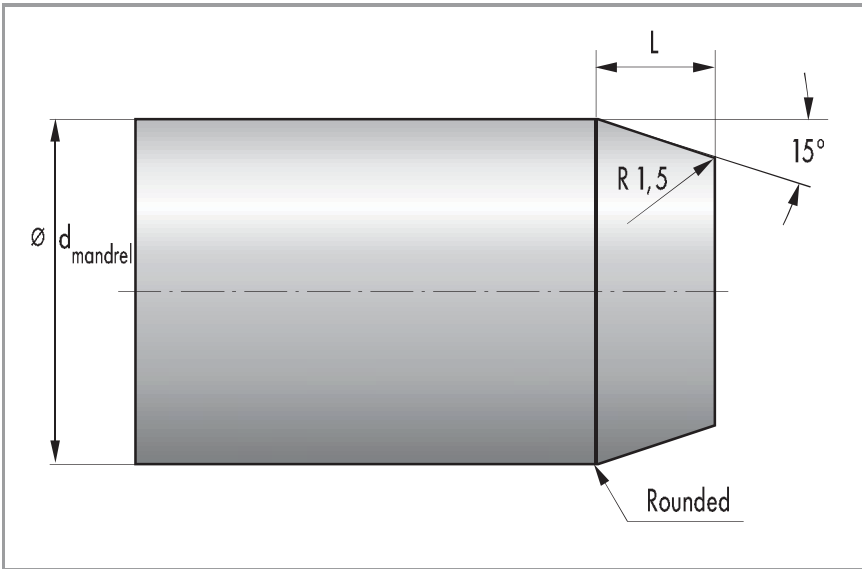


Fig. 50 Calibrating mandrel

Preferred materials: pressure mandrel – plastic
 tapered sleeve – plastic

We can supply complete assembly tools on enquiry.

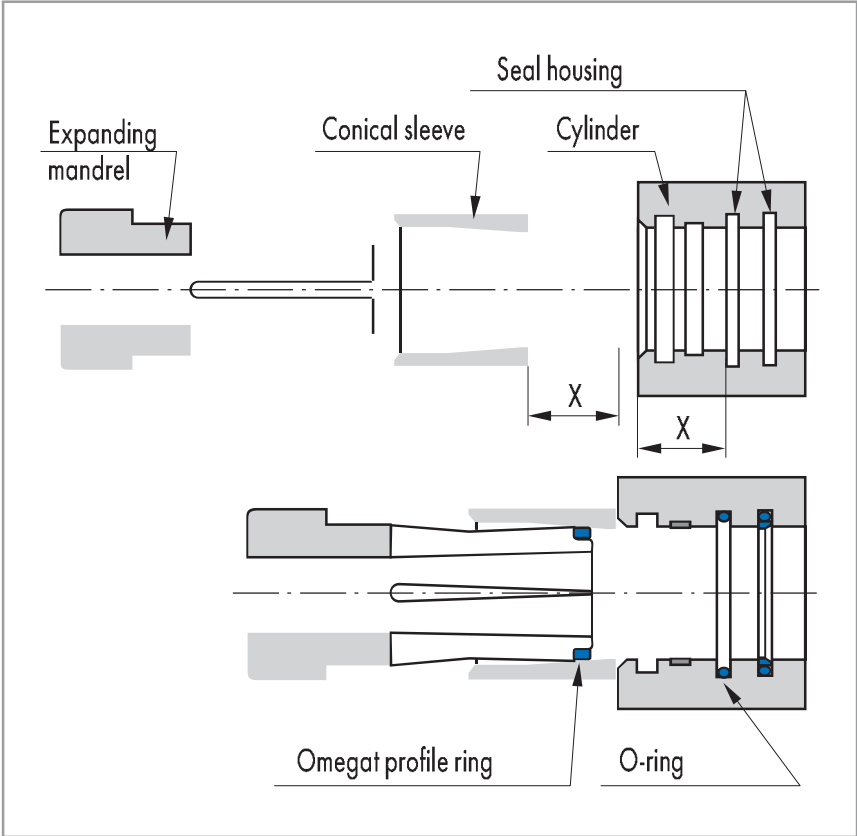
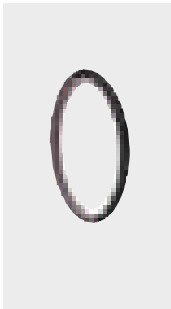


Fig. 51 Finished assembly tools (on enquiry)



Omegat Piston Seal

Omegat Piston Seals must always be fitted in single-piece pistons with a fitting tool.

Insert O-Ring into the groove without twisting it.

Stretch Omegat Profile Ring with spreader sleeve over a tapered mounting sleeve and spring into the groove, for larger dimensions use mounting belt (order no. 24346745) (no sharp-edged tools).

Calibrate Omegat Profile Ring on the piston diameter with a slip ring. When using profile rings with an L-dimension $\geq 6,3$ mm we recommend using a plastic tension belt.

Expanding mandrel approx. 5° chamfer

PTFE profile ring

Expanding sleeve

O-ring

Tbl. 11 Mounting equipment for piston seal

Fitting instructions for
Merkel Compact Seal L 43

Fitting of the Merkel Compact Seal L 43 is uncomplicated and generally corresponds to the conventional compact piston seal. The assembly should be carried out in the sequence below.

Fitting of Merkel Compact Seal T 19

We recommend using installation tools for all dimensions of the size series T 19; hand installation is possible for servicing. The sequence when mounting the individual parts is as follows:

- First angled bush
- Sealing component
- Second angled bush

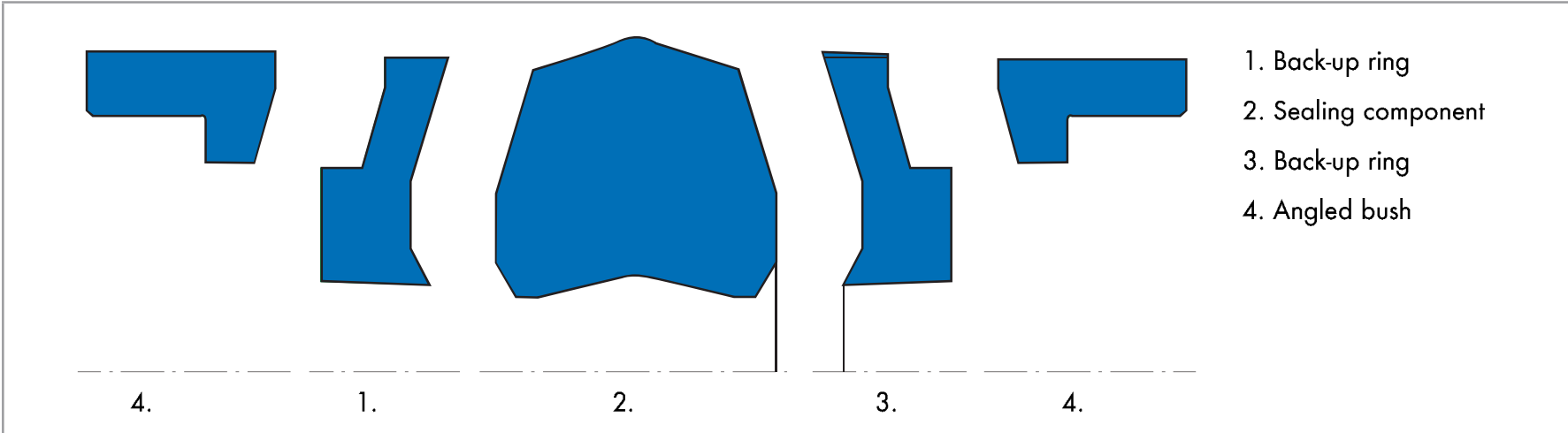
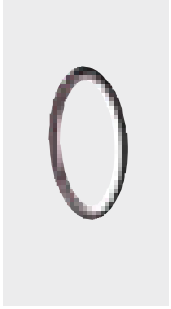


Fig. 52 Fitting of Merkel Compact Seal L 43



Installation instructions for Merkel double wiper PT 2

Double wipers of the PT 2 series can be installed in housings that are not axially accessible without tools from $\varnothing 150$ mm. For installation first insert the large O-ring into the groove, insert the small O-ring into the groove of the PTFE profile ring, then the profile ring is placed in a kidney shape and snapped in. Make sure that the profile ring is not kinked and that the sealing edge is correctly aligned to the pressure direction. For smaller dimensions please use an installation tool. Dimensions $< \varnothing 100$ mm cannot be installed in a plunge-cut groove.

Installation instructions for Merkel double wiper PT 1

Double wipers of the PT 1 series with inside diameter ≥ 30 mm can be installed in non-axially accessible housings without fitting tools. For smaller dimensions a fitting tool is recommended.

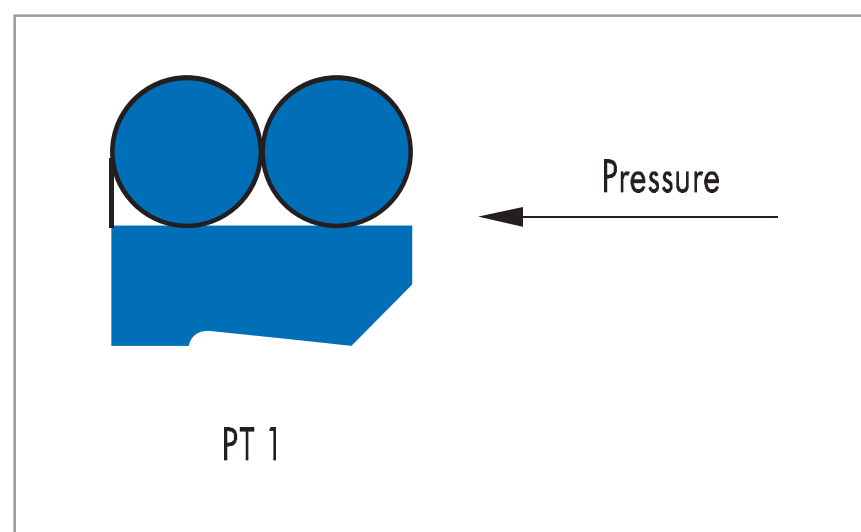


Fig. 53 Functional direction of wiper

For installation the O-rings are inserted into the groove, then the profile ring is placed in a kidney shape and snapped in. Make sure that the profile ring is not kinked and the sealing edge is correctly aligned to the pressure direction.

Fitting of Chevron Seal Sets

Instructions for Chevron Seal Set housings

Adjustable housings have the advantage of an optimal adjustment option. After a lengthy period of running and incipient wear on the seal tightening the gland can extend the service life and significantly delay a system standstill. For adjustable housings an extension of 2,5% and an adjustability of 7,5% of dimension L is recommended. Non-adjustable housings have the advantage of more cost-effective manufacture, because washers are not required. Seal Set Type B is particularly recommended for these housings. The rubber-sprung back-up rings handle the function of initial compression and continuous re-adjustment during operation. Maintenance of the seal contact area is not required.

Fitting

Before installation all individual parts of the seal set must be evenly greased. Mineral-oil-based greases can be used. The rod must be in the cylinder's installation space during fitting. The individual parts of the set must be installed separately in the chamber. In this regard make sure that the seals are not reversed. Open chevron seal sets are used for repairs, e.g. in large systems, if endless seals cannot be installed.

Please note

Open chevron seals have an oversize in the circumference length to allow sufficient compression and a good sealing effect at the joint sections. An endless delivered seal set should therefore not be cut. Open chevron seals are always supplied with installed profile cords.

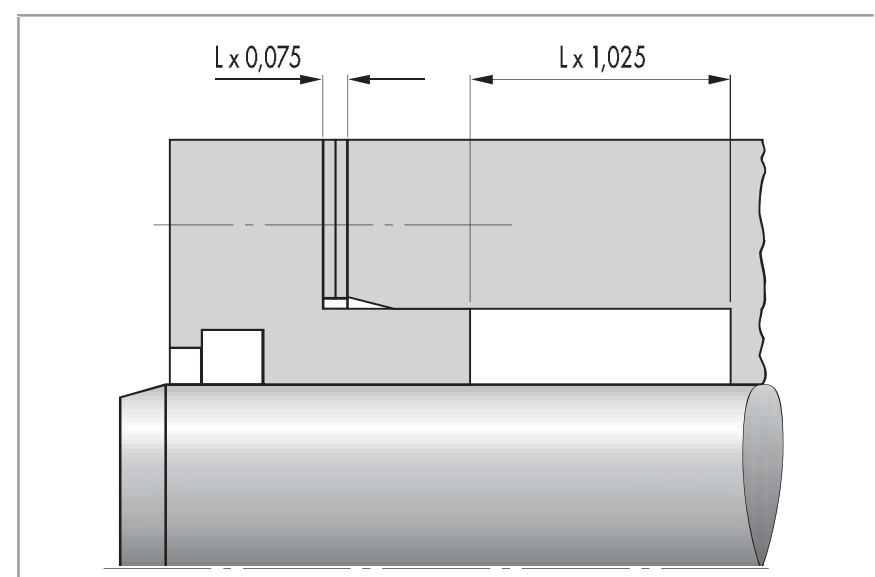
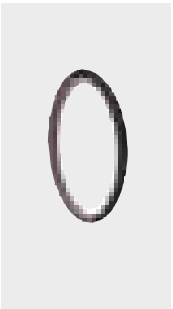


Fig. 54 DMS installation space example



Fitting of Merkel Forseal made of PTFE

We recommend installation of Merkel Forseal Seals in divided grooves.
In exceptional cases there is also the option of snap-in fitting in semi-open grooves. We request you consult us in these t applications.
The sealing ring must not be kinked during fitting.

Installation instructions for
Merkel Forseal FOI made of PTFE

Groove cross-section	O-Ring Ø	X _{min}	can be installed from FOI-	L
1,45 x 2,4	1,78	0,2	12	4,0
2,25 x 3,6	2,62	0,3	20	4,5
3,10 x 4,8	3,53	0,5	30	5,0
4,70 x 7,1	5,33	0,6	40	7,0
6,10 x 9,5	7,00	0,7	55	9,0

Tbl. 12 Rod housing

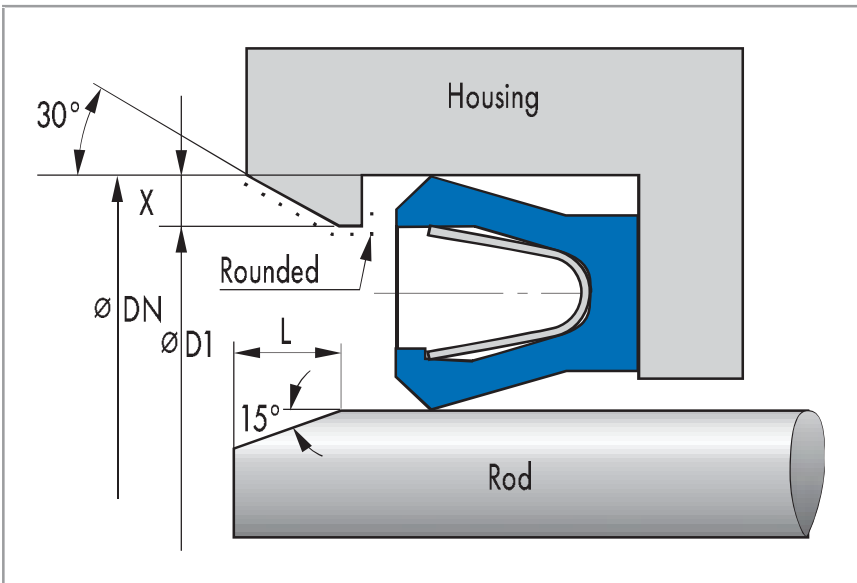


Fig. 55 Installation in half-open grooves/snap-in fitting
Merkel Forseal FOI

Installation instructions for
Merkel Forseal FOA made of PTFE

Groove cross-section	O-Ring Ø	X _{min}	can be installed from FOA-	L
1,45 x 2,4	1,78	0,4	15	4,0
2,25 x 3,6	2,62	0,6	20	4,5
3,10 x 4,8	3,53	0,7	25	5,0
4,70 x 7,1	5,33	0,8	30	7,0
6,10 x 9,5	7,00	0,9	45	9,0

Tbl. 13 Piston housing

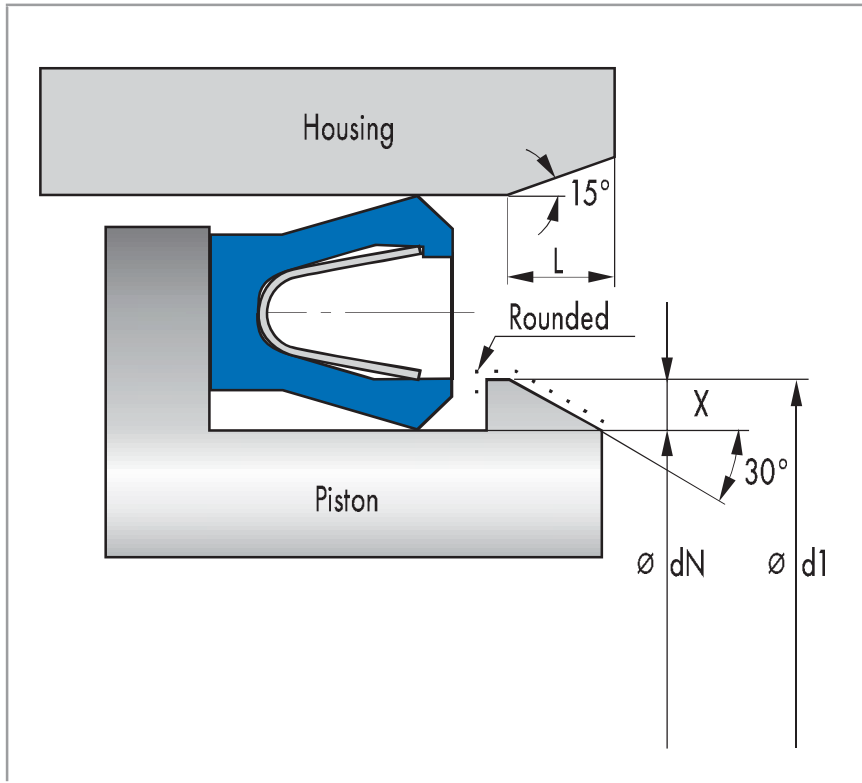


Fig. 56 Installation in half-open grooves/snap-in fitting
Merkel Forseal FOA